Assignment No 2

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Submitted to: Sir Asad
Subject: PF (Programing fundamental)
Topic: variable , Constant, Arithmetic Operator
Section: CS E
Department: CS and IT
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Task 1

Program 01

```
#include <iostream>
using namespace std;
int main(){
   char letter;
   letter = 'A';
   cout << letter << "\n";
   letter = 'B';
   cout << letter << "\n";
   return 0;
}
```

Output:

A
B

Program 02

```
#include <iostream>
using namespace std;
int main(){
   char letter;
   letter = 65;
   cout << letter << "\n";
   letter = 66;
   cout << letter << "\n";
   cin >> letter;
   return 0; }
```

Output

A
B

.

Program 03

```
#include <iostream>
Using namespace std;
int main() {
int letter;
Letter = 'A';
Cout << Letter <<"\n",
Letter = 'B';
```

Cout <<< letter << “\n”,
Cin >> letter;
return 0; }

Output:
65
68.

Task #2

Question:
Take two integers as input from the user. Apply arithmetic operations on them (+, -, *, /) and print them on screen.
Program:

```
#include <iostream>
using namespace std;

int main()
{
   int num1, num2;

   cout << “Enter first integer: “;
   cin >> num1;

   cout << “Enter second integer: “;
   cin >> num2;

   cout << “Addition = “ << num1 + num2 << endl;
   cout << “Subtraction = “ << num1 - num2 << endl;
   cout << “Multiplication = “ << num1 * num2 << endl;
   cout << “Division = “ << num1 / num2 << endl;

   return 0;
}
```

Output:

```
Enter first integer: 20
Enter second integer: 73

Addition = 93
Subtraction = -53
Multiplication = 1460
Division = 0
```

Task 03

Write the output of the following Program

```
#include <stdio.h>
#include <iostream>
using namespace std;

int main() {
```

```
   int val1;
   int val2;
   int val3;

   cout << “Please enter your 3 numbers:”;
   cin >> val1 >> val2 >> val3;

   cout << val1 << “\n” << val2 << “\n” << val3 << “\n”;
   cout << val3 << “\n” << val2 << “\n” << val1 << “\n”;

   return 0;
}
```

Output

```
Please enter your 3 numbers: 10
40
50

10
40
50
50
40
10
```

Task 4:

Write a C++ program to calculate the distance between the two points.

Note: x1, y1, x2, y2 are all double values.

Formula:

```
#include <iostream>
#include <cmath> using namespace std;
int main()
double x1, y1, x2, y2, d;
cout << “Enter x1: “; cin >> x1;
cout << “Enter y1: “; cin >> y1;
cout << “Enter x2: “; cin >> x2;
cout << “Enter y2: “; cin >> y2;
d = sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
cout << “Distance = “ << d << endl;
return 0;
}
```

Output

```
Enter x1: 2
Enter y1: 3
Enter x2: 5
Enter y2: 6
Distance = 4.24264
```

Task 5:

1. Calculate the Area of a Circle
2. Calculate the Area of a Rectangle
3. Calculate the Area of a Triangle.

```
#include <iostream>
using namespace std;
int main()
{
// 1. Area of Circle
float r = 5;
float circle = 3.14 * r * r;

// 2. Area of Rectangle
float length = 10, width = 5;
float rectangle = length * width;

// 3. Area of Triangle
float base = 8, height = 6;
float triangle = 0.5 * base * height;
cout << “Area of Circle = “ << circle << endl;
cout << “Area of Rectangle = rectangle << endl; <<
11 cout << “Area of Triangle = “ << triangle << endl;
return 0;
}
```

Output

```
Area of Circle 78.5
Area of Rectangle = 50
Area of Triangle = 24
```

Task-6

Total Purchase A customer in a store is purchasing five items. The price of the items five are ?

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
   double item1 = 12.95;
   double item2 = 24.95;
   double item3 = 6.95;
   double item4 = 14.95;
   double item5 = 3.95;

   double subtotal;
   double salesTax;
   double total;

   subtotal = item1 + item2 + item3 + item4 + item5;
   salesTax = subtotal * 0.06;
   total = subtotal + salesTax;

   cout << fixed << setprecision(2);

   cout << “Price of item 1 = $” << item1 << endl;
   cout << “Price of item 2 = $” << item2 << endl;
```

```
    cout << “Price of item 3 = $” << item3 << endl;
    cout << “Price of item 4 = $” << item4 << endl;
    cout << “Price of item 5 = $” << item5 << endl;

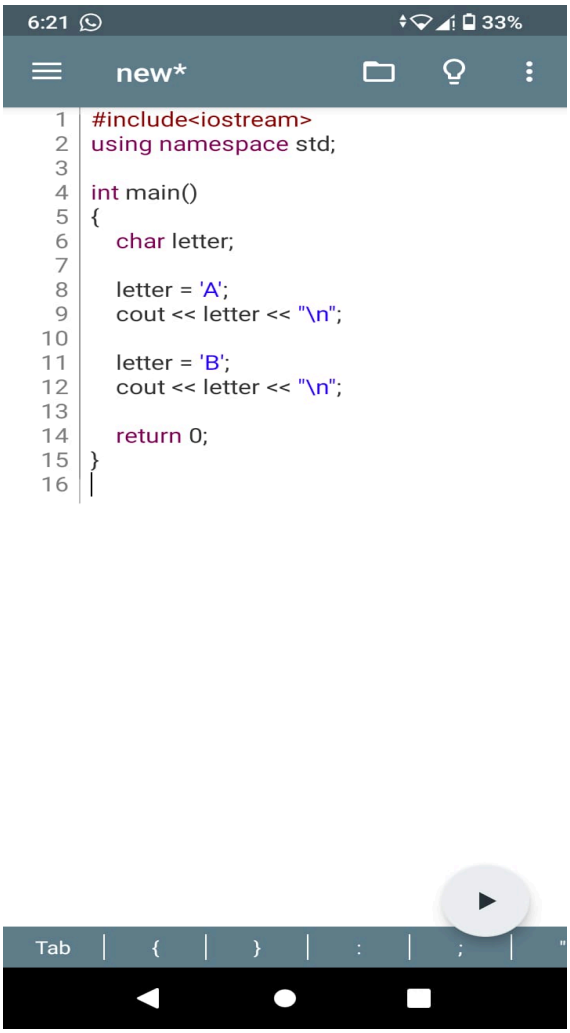
    cout << “Subtotal = $” << subtotal << endl;
    cout << “Sales Tax = $” << salesTax << endl;
    cout << “Total = $” << total << endl;

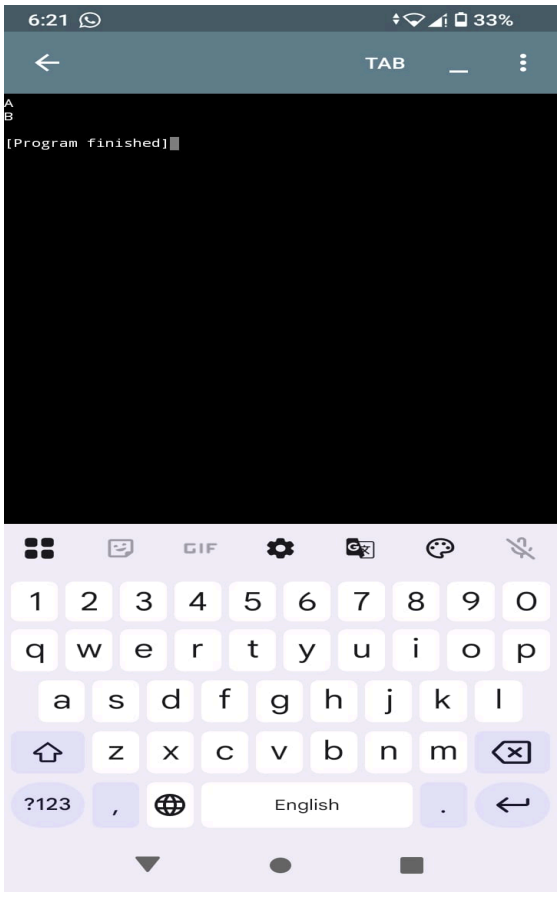
    return 0;
}
```

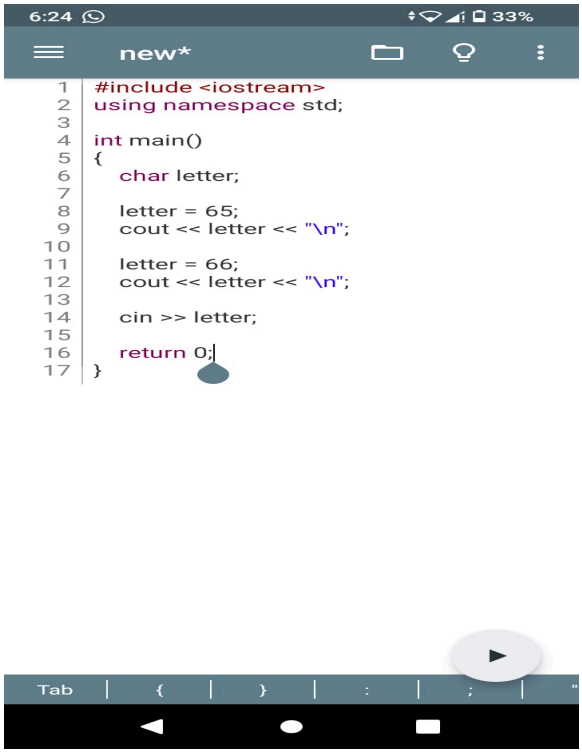
Output:

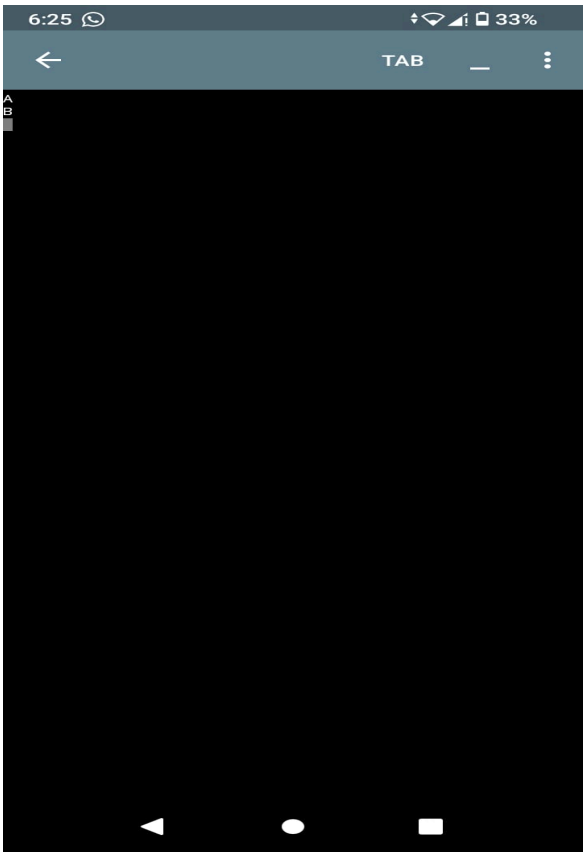
```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.83
Total = $67.58
```

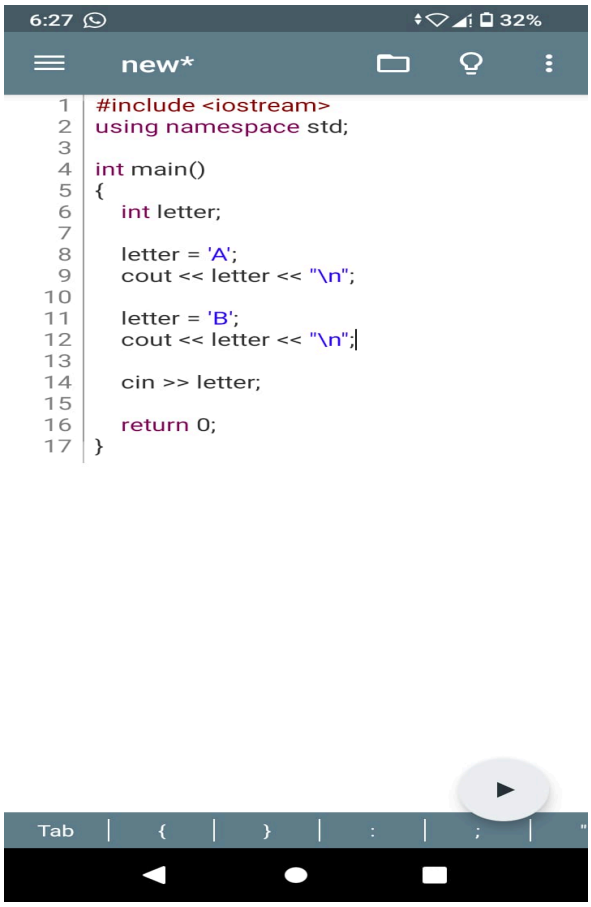
.....

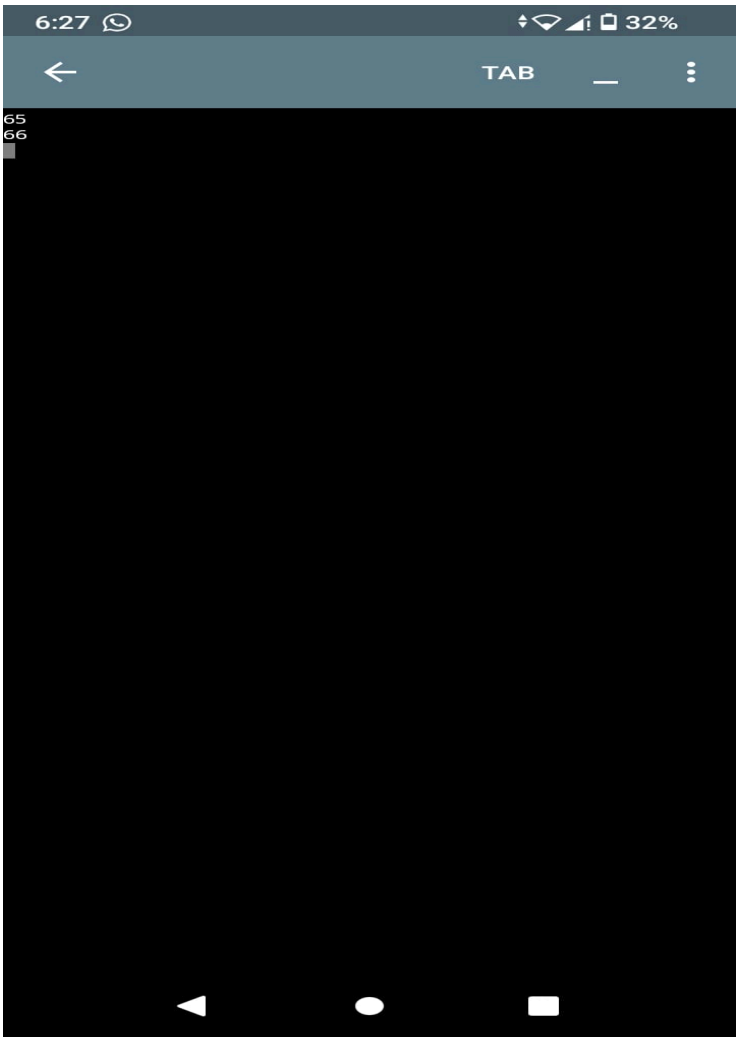


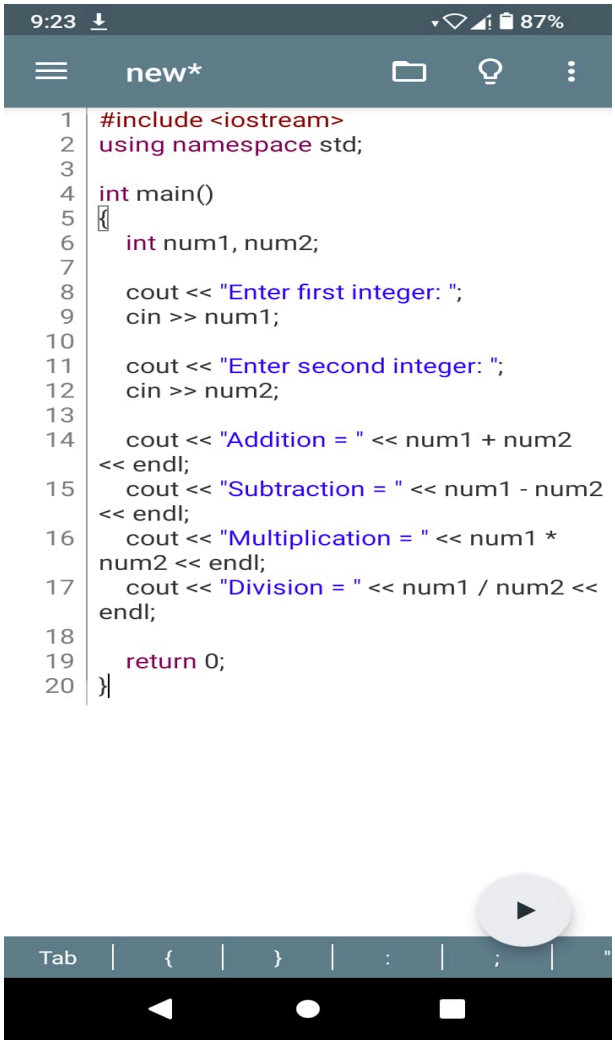


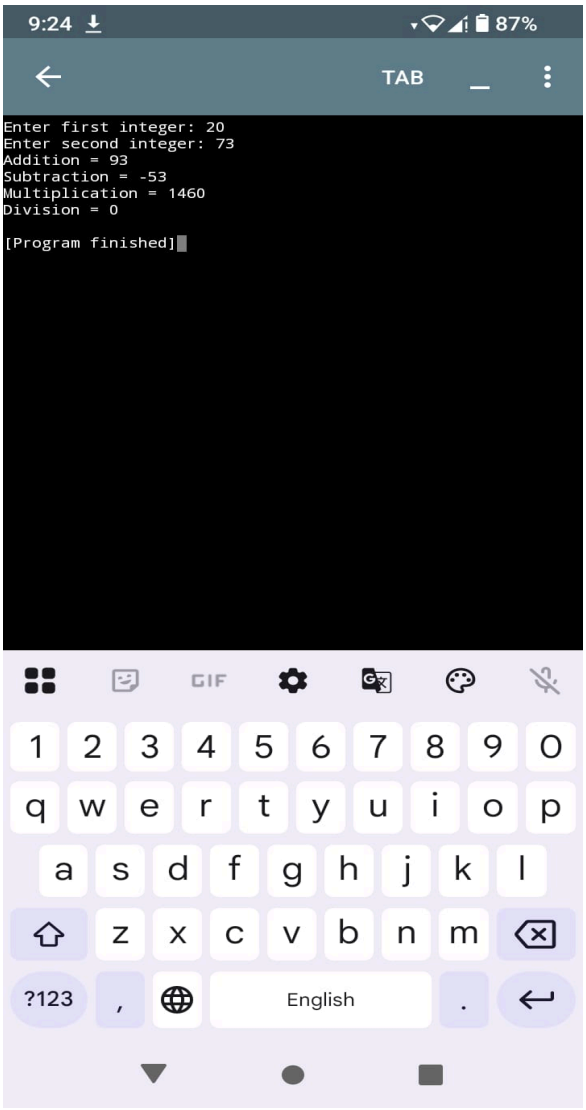

6:24
33%
new*
#include <iostream>
using namespace std;
int main()
{
char letter;
letter = 65;
cout << letter << "\n";
letter = 66;
cout << letter << "\n";
cin >> letter;
return 0;
}
Tab
{
}
:
;

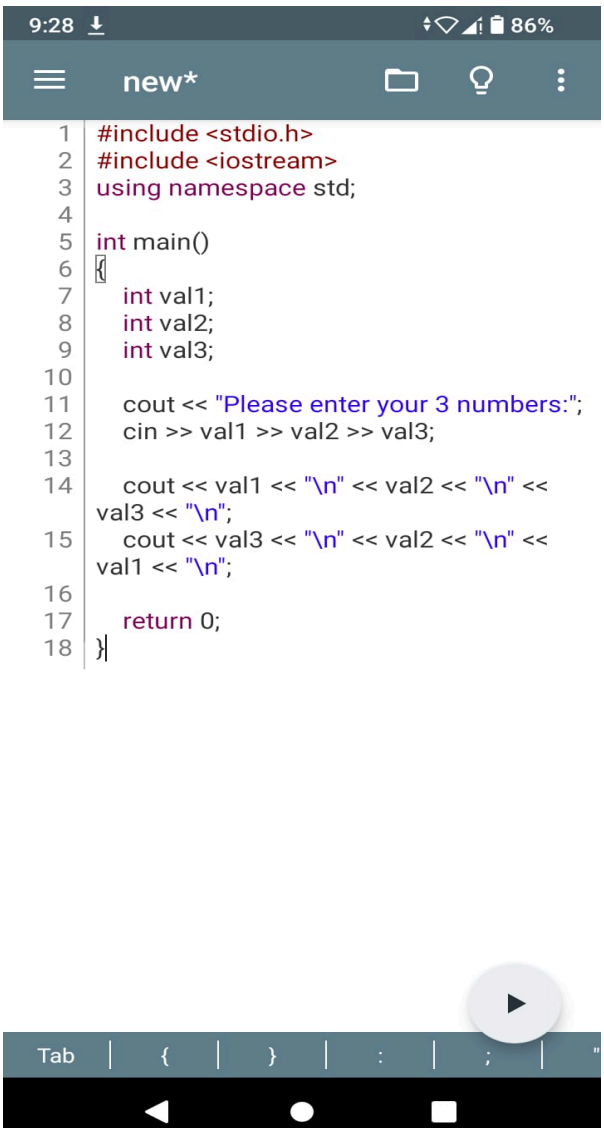

6:25
33%
TAB
A
B


6:27
32%
new*
#include <iostream>
using namespace std;
int main()
{
int letter;
letter = 'A';
cout << letter << "\n";
letter = 'B';
cout << letter << "\n";
cin >> letter;
return 0;
}
Tab
{
}
:
;

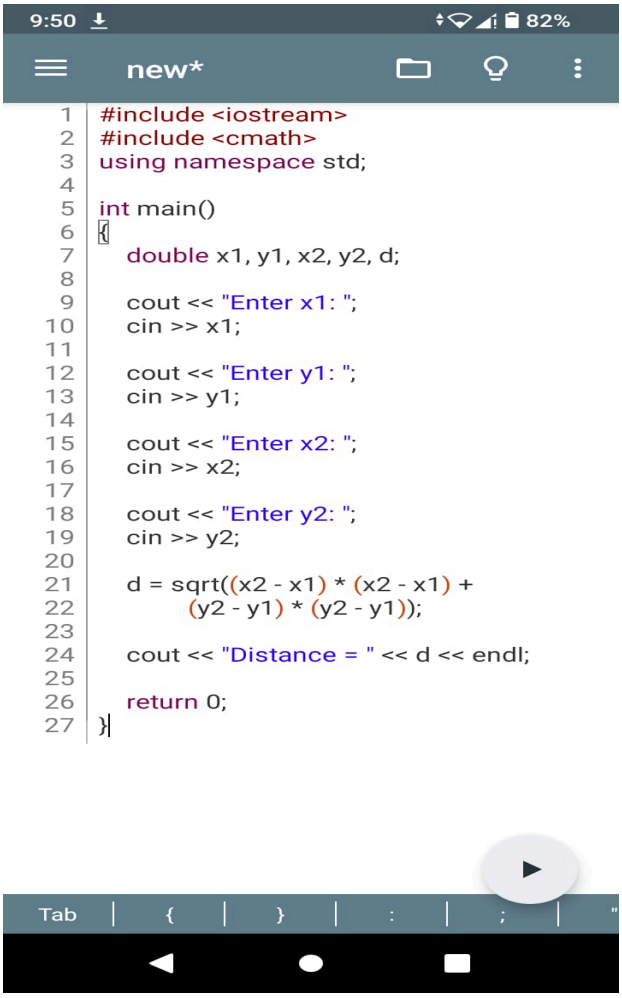

6:27
32%
TAB
65
66

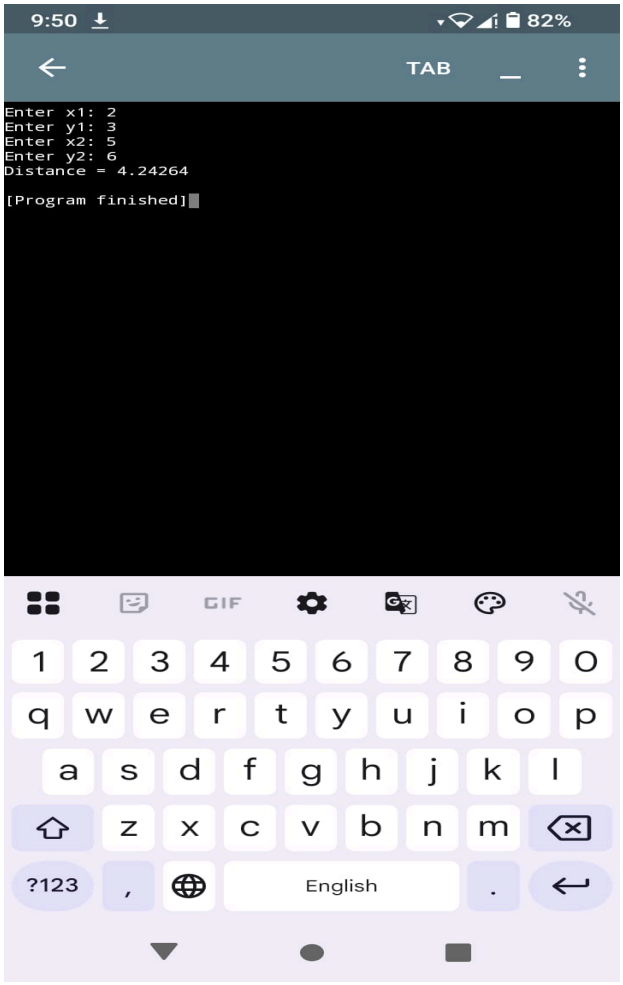

9:23
87%
new*
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6 int num1, num2;
7
8 cout << "Enter first integer: ";
9 cin >> num1;
10
11 cout << "Enter second integer: ";
12 cin >> num2;
13
14 cout << "Addition = " << num1 + num2
<< endl;
15 cout << "Subtraction = " << num1 - num2
<< endl;
16 cout << "Multiplication = " << num1 *
num2 << endl;
17 cout << "Division = " << num1 / num2 <<
endl;
18
19 return 0;
20 }
Tab { } : ;

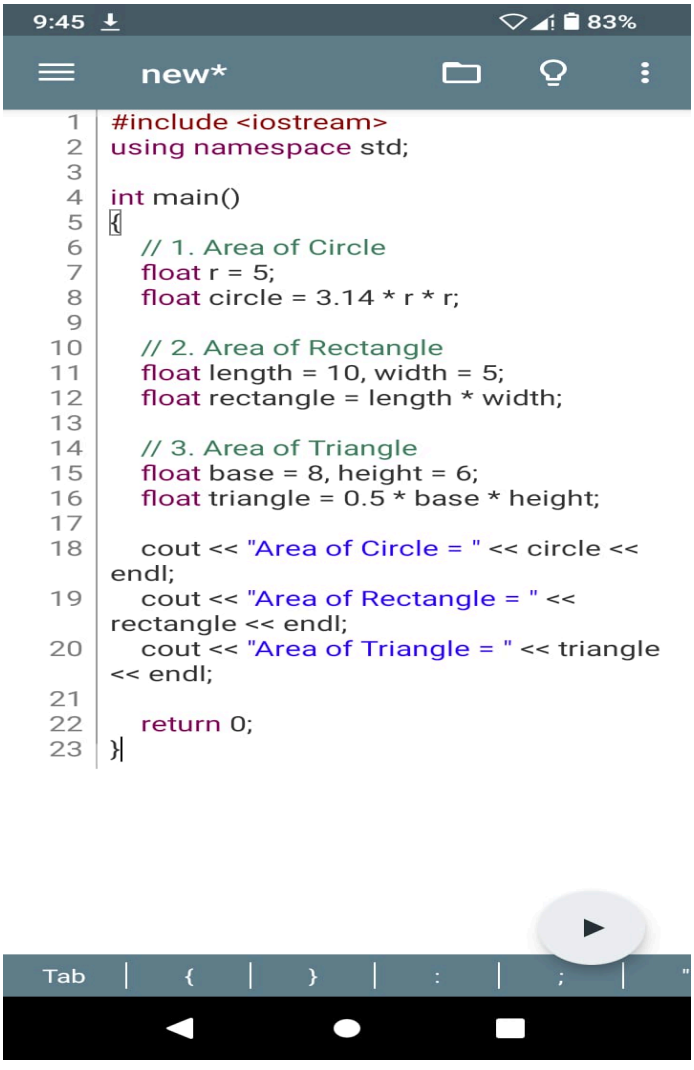

9:24
87%
TAB
Enter first integer: 20
Enter second integer: 73
Addition = 93
Subtraction = -53
Multiplication = 1460
Division = 0
[Program finished]
GIF
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123 , English .

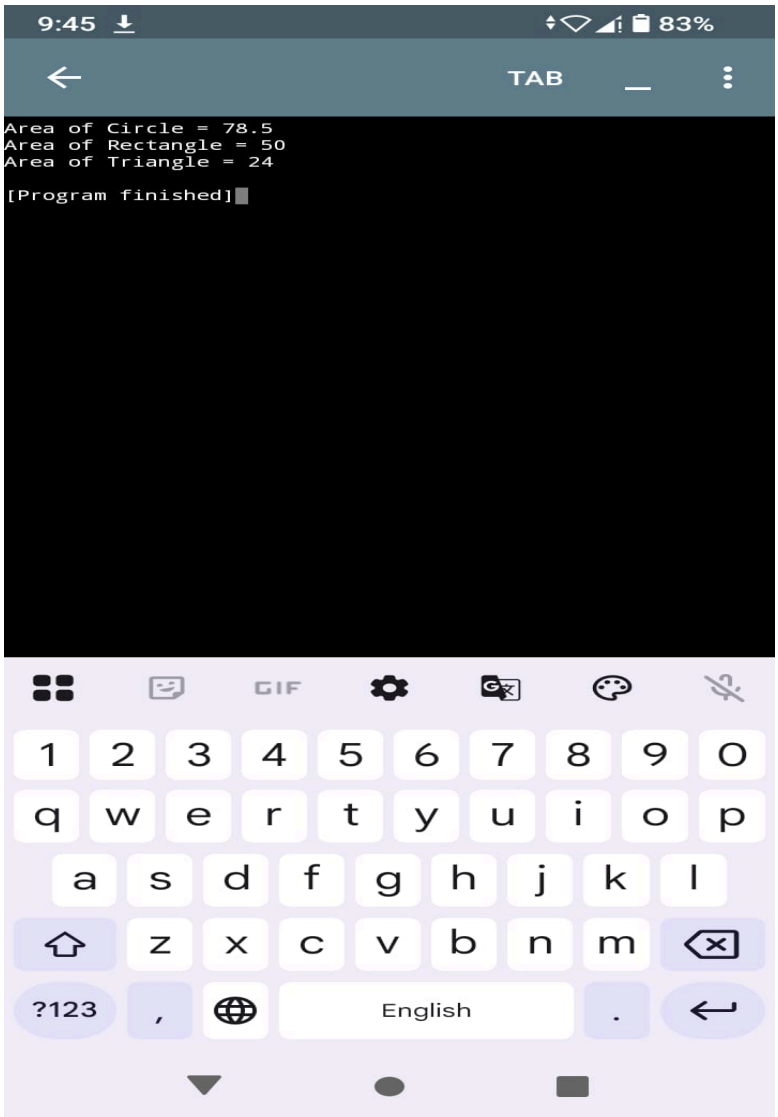

9:28
86%
new*
1 #include <stdio.h>
2 #include <iostream>
3 using namespace std;
4
5 int main()
6 {
7 int val1;
8 int val2;
9 int val3;
10
11 cout << "Please enter your 3 numbers:";
12 cin >> val1 >> val2 >> val3;
13
14 cout << val1 << "\n" << val2 << "\n" <<
val3 << "\n";
15 cout << val3 << "\n" << val2 << "\n" <<
val1 << "\n";
16
17 return 0;
18 }
Tab { } : ;

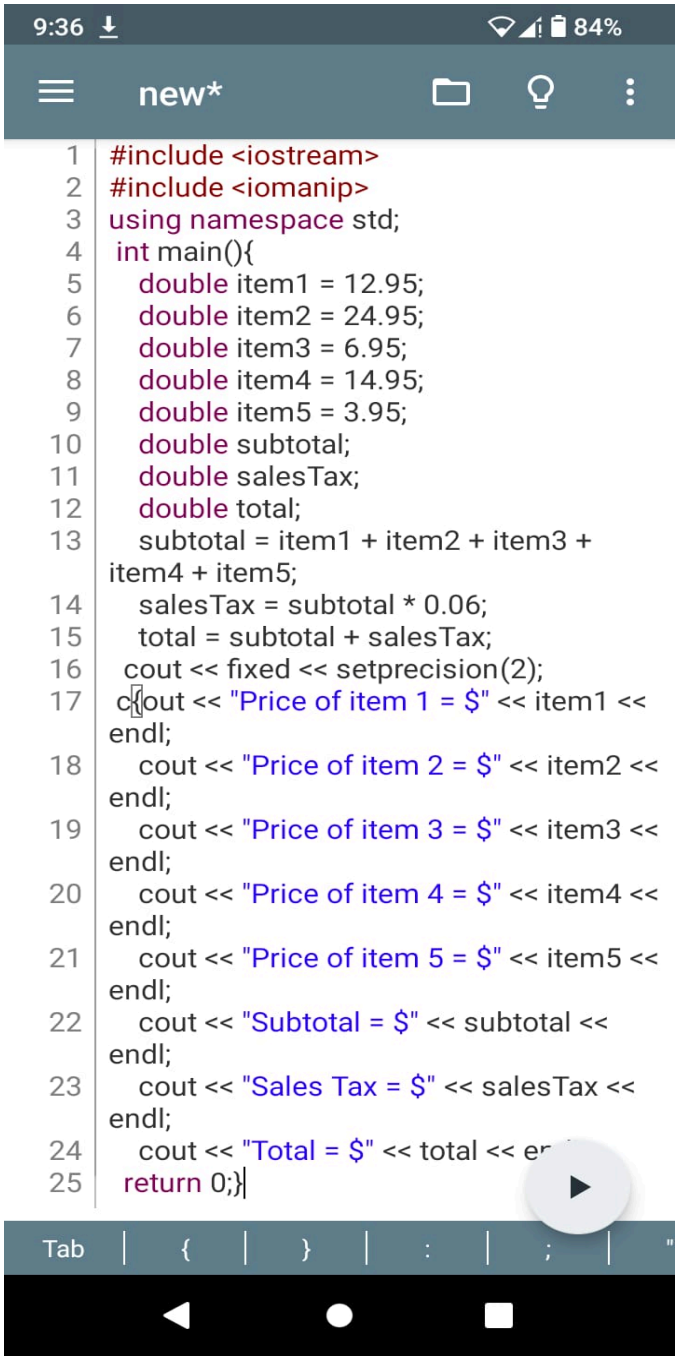

9:28
86%
TAB
Please enter your 3 numbers:10
40
50
10
40
50
50
40
10
[Program finished]
GIF
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123 , English .

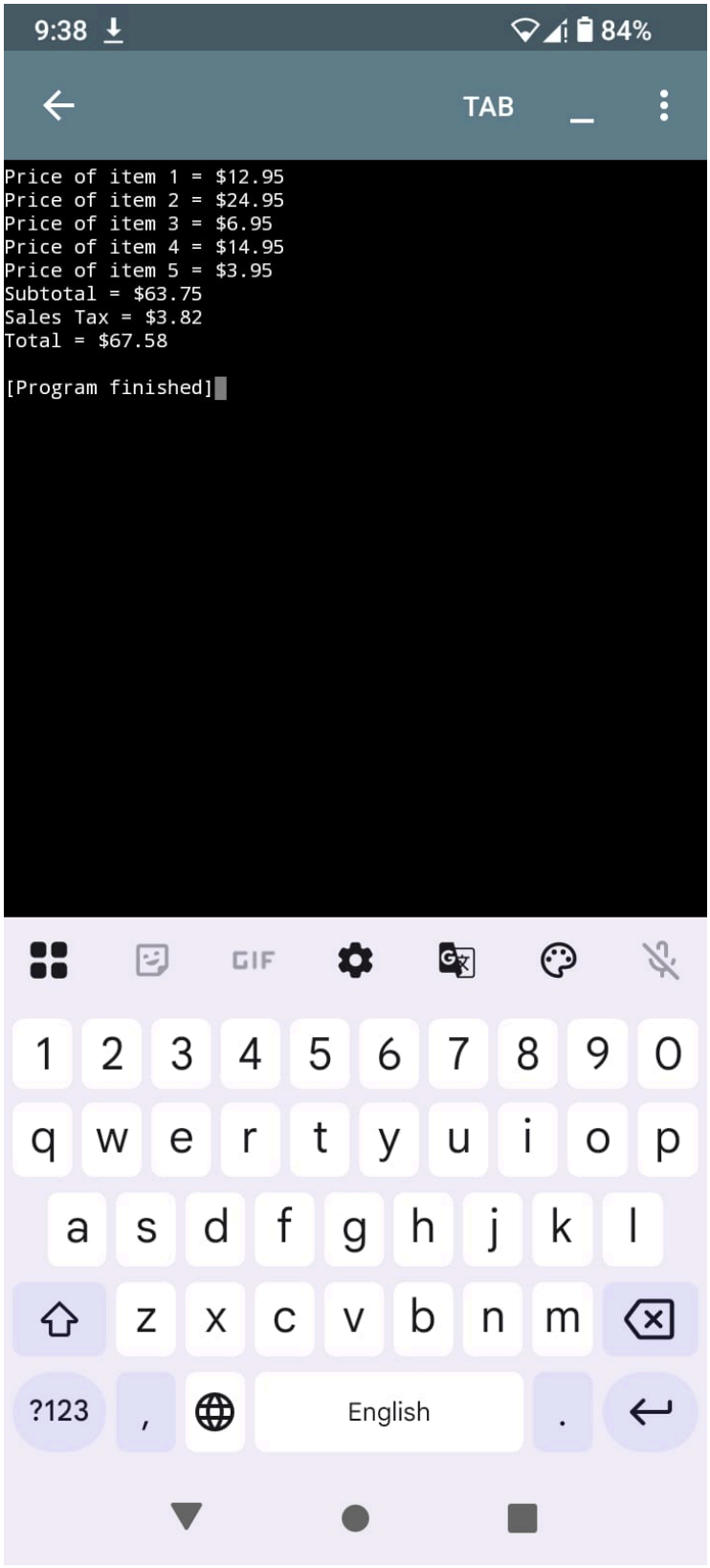

9:50
82%
new*
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
double x1, y1, x2, y2, d;
cout << "Enter x1: ";
cin >> x1;
cout << "Enter y1: ";
cin >> y1;
cout << "Enter x2: ";
cin >> x2;
cout << "Enter y2: ";
cin >> y2;
d = sqrt((x2 - x1) * (x2 - x1) +
(y2 - y1) * (y2 - y1));
cout << "Distance = " << d << endl;
return 0;
}
Tab { } : ; "


9:50
82%
TAB
Enter x1: 2
Enter y1: 3
Enter x2: 5
Enter y2: 6
Distance = 4.24264
[Program finished]
GIF
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123 , English .


9:45
83%
new*
#include <iostream>
using namespace std;
int main()
{
// 1. Area of Circle
float r = 5;
float circle = 3.14 * r * r;
// 2. Area of Rectangle
float length = 10, width = 5;
float rectangle = length * width;
// 3. Area of Triangle
float base = 8, height = 6;
float triangle = 0.5 * base * height;
cout << "Area of Circle = " << circle << endl;
cout << "Area of Rectangle = " << rectangle << endl;
cout << "Area of Triangle = " << triangle << endl;
return 0;
}
Tab { } : ; "


9:45
83%
TAB
Area of Circle = 78.5
Area of Rectangle = 50
Area of Triangle = 24
[Program finished]
GIF
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123 , English .


9:36
84%
new*
#include <iostream>
#include <iomanip>
using namespace std;
int main(){
double item1 = 12.95;
double item2 = 24.95;
double item3 = 6.95;
double item4 = 14.95;
double item5 = 3.95;
double subtotal;
double salesTax;
double total;
subtotal = item1 + item2 + item3 + item4 + item5;
salesTax = subtotal * 0.06;
total = subtotal + salesTax;
cout << fixed << setprecision(2);
cout << "Price of item 1 = $" << item1 << endl;
cout << "Price of item 2 = $" << item2 << endl;
cout << "Price of item 3 = $" << item3 << endl;
cout << "Price of item 4 = $" << item4 << endl;
cout << "Price of item 5 = $" << item5 << endl;
cout << "Subtotal = $" << subtotal << endl;
cout << "Sales Tax = $" << salesTax << endl;
cout << "Total = $" << total << endl;
return 0;}
Tab
{
}
:
;


9:38
84%
TAB
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.82
Total = $67.58
[Program finished]
GIF
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123
English